



Empathic conversational agents for real-time monitoring and co-facilitation of patient-centered healthcare

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ABSTRACT

Healthcare systems across the world are transitioning into patient-centered healthcare models to ensure improved health outcomes, increased operational efficiencies and respectful patient engagement. Digital health technologies are at the forefront of this transition in facilitating a role for the patient in the clinical dimensions of the healthcare trajectory, from diagnosis and interventions to treatment and recovery. Despite this prevalence in the clinical space, the non-clinical needs of patient mental health and wellbeing are frequently overlooked by contemporary patient-centered healthcare models. Conversational agents (or chatbots) are digital dialogue systems that are widespread and widely used in sequential information provision and information acquisition tasks. Given the intimate nature of this human-machine interaction, conversational agents can be effectively utilized to support and sustain patient mental health and wellbeing. In this paper, we propose an empathic conversational agent framework based on an ensemble of natural language processing techniques and artificial intelligence algorithms for real-time monitoring and co-facilitation of patient-centered healthcare for improved mental health and wellbeing outcomes. The technical contributions of this framework are; detection of patient emotions, prediction of patient emotion transitions, detection of group emotions, formulation of patient behavioral metrics, and resource recommendations based on patient concerns. The architectural contributions of the framework are intelligent communication channels that stream empathic conversational elements and resource recommendations for the multi-user conversations and co-facilitation updates for the human healthcare provider interface. The framework was empirically evaluated on a benchmark dataset and further validated based on a clinical protocol designed for its application in an online support group setting for cancer patients and caregivers in Canada. The results of these experiments confirm the effectiveness of this framework, its contributory role and practical value in realizing a patient-centered healthcare model for improved mental health and wellbeing outcomes.

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1. Introduction

Patient-centered healthcare has sustained a steady progression to acceptance from its conception in the late 1960s, when it was introduced as patient-centered medicine, in contrast to illness-oriented medicine, by Enid Balint [1]. The Institute of Medicine formalized this effort by defining patient-centered care (PCC) as “health care that establishes a partnership among practitioners, patients, and their families (when appropriate) to ensure that decisions respect patients’ wants, needs, and preferences

and that patients have the education and support they need to make decisions and participate in their own care”, and identified it as one of six core elements of high-quality care [2,3]. There have also been contrary efforts to commoditize PCC as consumer-oriented care, and frame it as a free market that is driven by demand and supply [4]. However, digital health technologies have been central to the wider promotion and adoption of PCC, with evidence indicating access to information, patient activation, and care coordination leading to improved health outcomes. Digital health records, digital decision aids, mobile health apps, telehealth, virtual reality tools, agents and bots are some of the digital health technologies and applications widely used in PCC [5].

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Given this increased emphasis on the physical and clinical dimensions of the health trajectory for PCC, there has been limited focus on the research and development of digital health technologies for the mental health and wellbeing of patients. This underexplored dimension of PCC is becoming vital as patients are more informed, connected and socially aware of their medical condition and prognosis.

The use of technology is becoming increasingly popular and pertinent in capturing mental health issues in patients. Signals generated via wearable devices [6], ubiquitous sensors [7] and even social media [8,9] are becoming useful given their practical implications. In this sphere, the use of conversational agents is deemed as a necessity as well as a crucial component in digital communication. Conversational agents (chatbots) aim to automate communicative labor, in both oral and written conversations [10]. The mainstream success of voice assistance chatbots such as Siri, Alexa, Cortana and Google [11] are either Q&A based or serve a well-defined repetitive or sequential purpose of information acquisition/provision. More recent research demonstrate the use of advanced technology to build open-domain chatbots that mimic human-like conversations [12,13]. This shows the importance and the usability of using chatbots in streamlining and automating communication in wide array of applications. In the health domain, previous research studies indicate the use of chatbots as counseling applications which are immensely useful to enhance patient communication using digital platforms [14, 15]. Specifically, automating post-treatment communications and support groups have multiple benefits as a streamlined, continuous flow of communication can be established with the use of chatbot applications.

However, these research endeavors mostly focus on the question-answering mechanism and have less focus on the use of Natural Language Processing (NLP) techniques and Artificial Intelligence (AI) algorithms for the real-time detection of patient emotions and indicative behaviors such as group dynamics and thematic contexts. Furthermore, the real-time detection of emotions and behaviors is actionable as it leads to an effective understanding of patient needs and expectations, thereby improved mental health and wellbeing outcomes. In this paper, we propose an empathic conversational agent framework based on an ensemble of NLP and AI algorithms for real-time monitoring and co-facilitation of patient-centered healthcare. This is designed to provide automated responses as well as capture emotion behaviors of patients within a group setting and inform the human moderators regarding patients' emotion changes. Within the context of this framework, the key research contributions are;

(1) An ensemble of NLP and AI algorithms for (a) patient emotions analysis based on the detection of a spectrum of emotions and prediction of emotion transitions, (b) group emotion detection based on individual emotions and collaboration metrics, (c) patient behavioral changes based on active, passive participation and information, emotional support metrics, (d) resource recommendation based on medical and social concerns.

(2) An intelligent filtering approach for the propagation of empathic conversational elements and co-facilitation updates for the human therapist or care provider.

(3) Validation of this framework based on a clinical protocol designed for its application on an online, professionally led, synchronous, text-based online support group for cancer patients and caregivers across six provinces in Canada [16].

The rest of the paper is organized as follows. Section 2 presents related work on patient-reported information, patient emotions, online support groups and the use of chatbots for PCC. Section 3 delineates the methodological novelty and development of the proposed empathic conversational agent framework for real-time monitoring and co-facilitation of patient-centered healthcare. Section 4 reports experiments and results, and Section 5 concludes the paper.

2. Related work

With the emergence of digital communication platforms, patients are readily engaging in online conversations related to their health concerns [17]. Most of the patients engage in support groups which facilitate the sharing of concerns, thoughts and their emotional burden with others. These groups are often moderated by a therapist or an accredited member. Online support has become a hallmark of PCC, given its anonymity, equitable accessibility to a broad population, and most importantly patients are able to openly express their opinions and emotions which is impractical in clinical settings. [18].

Online forums dedicated to different illnesses have gained the attention of patients with similar concerns, and act as a platform to share their concerns and emotions with peers. Subsequently, this process generates an enormous amount of patient-reported information which is of utmost importance in understanding patient behaviors and expectations [19]. The conversations are rich with patient emotions, their concerns and thoughts which are crucial to elevating the current PCC applications. The emotion expressed in online groups can be used to investigate different patient behavior patterns [20] and even use for clinical studies to assess the perception of different treatment types [21,22]. As the customization and individualization of PCC come from the patient-reported information, it is essential to investigate the emotions and concerns expressed in these conversations.

Furthermore, current literature reports diverse PCC applications including the use of conversational agents and assistants for the detection of depression and distress in conversations thereby provide emotional support [15,23]. It is apparent that the advent of new technologies brings the possibilities to elevate PCC using chatbots given the proliferation of content and mobile application settings [24,25]. A recent review of mental health chatbots, outlines while there are promising avenues in using chatbots for clinical psychology, given the heterogeneity of the content, more thorough investigations should be carried out to assess the effectiveness of chatbots [26].

This has led to research and development of empathic conversational agents to be used across a wide array of applications. Microsoft introduced the social bot, Xiaolce, which identifies topics in conversations, user intent and several positive and negative emotions [27]. Several research studies have employed empathic conversation datasets to train chatbot frameworks [28,29], that are then able to generate empathic responses. However, one of the challenges in this approach would be that the responses generated are specific to the training dataset hence will have limitations in adopting to several domains. This requires the need for identifying emotional expressions regardless of the training dataset or the domain. Once the emotions are detected in conversations, necessary actions can be programmed into the chatbot for customizability. In this study, we present an approach using an unsupervised emotion modeling that can be embedded into the generic question-answering flow of a healthcare chatbot.

3. An empathic conversational agent framework

The proliferation of chatbots across a number of domains for professional and personal use has triggered an interest towards the incorporation of 'humanness and emotions' into chatbots and their conversations [30–33]. Incorporating emotions begins with the detection and extraction of emotion from free-flowing conversations. Emotion extraction using supervised machine learning and prelabelled data has been researched for interactive dialogue systems. In [34], conversational elements are analyzed against a database of emotion words to extract the emotions expressed and each conversation element has been labeled with the expressed emotion. In [31], a chatbot application is trained with

movie dialogue data where a linguistic analysis and a classification approach are taken to extract and label emotional expressions. Furthermore, a deep learning based approaches have been adopted where convolutional neural networks and word embeddings have been used to detect sentiments of conversations [35, 36]. However, supervised emotion detection relies on labeled data which is a challenge in real-time conversation environments where the data is unstructured and unlabeled. Furthermore, these approaches are specifically focused on extracting emotions from textual data and labeling each conversation element with identified emotions therefore lack comprehensive modeling of emotions to identify individual characteristics of a patient through the development of an emotion profile. Separately, the extension of individual emotion to a group setting as group emotion, as well as the volume of information exchanged in a conversation based on active/passive participation and the information-seeking behaviors revealed are indicative and reflective of humanness in human-machine interaction [37,38]. Furthermore, the proposed chatbot framework can be customized and aligned to a clinical setting based on clinical protocols. Unlike generic chatbots, this alignment to real-world healthcare setting enables the chatbot to support clinicians and therapists with co-facilitation.

The proposed empathic conversational agent framework is a manifestation of these ideals of ‘humanness and emotion’ for the functional purpose of real-time monitoring and co-facilitation of PCC (Fig. 1). The framework consists of four technical capabilities, patient emotion analysis, group emotions, patient behavioral metrics and response generation. The following subsections describe each capability, its constituents and the novel ensemble of NLP and AI algorithms enabling each capability which contributes towards patient-centric automated communication.

3.1. Patient emotion analysis

Firstly, emotions are extracted from the patient’s conversation elements and an emotion sequence is generated. Secondly, the generated emotion sequence is input into the emotion state transition model. Thirdly, the probabilities of emotion change are analyzed to identify distinct clusters of individuals based on their emotional patterns. Fourthly, an emotion prediction module is created using these emotion transition probabilities.

3.1.1. Emotion sequence extraction

The emotion feature extraction is based on the psychological emotion model proposed by Plutchik which consists of eight basic emotions; *anger, fear, sadness, disgust, joy, surprise, trust, anticipation* [39]. Based on this psychological model, we created a domain-specific emotion vocabulary, using several pretrained language models.

The emotion expression mining was carried out using an ensemble of two word embedding models. A Word2vec model was trained based on the datasets to capture emotion expressions related to the dataset and a pre-trained embedding model; Glove [40] was used to capture out-of-context emotion expressions.

First, we used the word2vec [41] embedding model in the continuous bag of words (CBOW) configuration. In a word embedding, words are represented by dense vectors where each vector denotes the projection of the word into continuous vector space. We used real-world patient conversations to train the word embedding model. Once the model was trained, it was queried to generate similar terms for each emotion. We trained two word2vec models, one using single word tokens and another model using bigrams. The use of two models enhanced the capability to capture more contextual information, in forms of single emotion terms as well as phrases that were present in the dataset.

Next, GloVe, another pre-trained model was used to query for out of context emotion expressions. Glove is pretrained on Wikipedia and Gigaword 5 dataset [42]. The main objective of using such models trained on larger datasets is to capture the emotion terms that are not present in the dataset used in the experiment therefore will capture more scope and context in emotion extraction.

The two models were then queried to generate similar emotion terms to each of the basic emotion. Similar terms were generated by computing cosine similarity between a mean of the projection weight vectors of the emotion term, and the vectors for each word in the model.

The following example shows an excerpt from the similar words generated by the trained word2vec model and the Glove model for the emotion ‘anger’.

Word2Vec model: anger = {“furious”, “angry”, “unfairness”, “unresponsive”, “irritated”, “annoyed”, ...}

Glove model: anger = {‘frustration’, ‘outrage’, ‘resentment’, ‘stirred’, ‘dismay’, ‘hostility’, ‘provoking’, ...}

Apart from the generic terms related to emotion ‘anger’, the word2vec model has picked terms such as “unfairness” and “unresponsive” which are used by patients and are associated with the term ‘anger’. This enabled capturing contextual emotional expressions from data and such terms captured were then added to generate specific ontology of emotional expressions for each emotion category. The querying process was conducted for all basic emotions and the top 50 similar terms of each model were taken to develop the emotion ontology. Moreover, this querying process can be used to extract emotion terms related to any sub-category of emotions. For example, it is also possible to incorporate additional emotions (‘stress’, ‘frustration’, ‘loneliness’ etc.) to customize based on the domain in which the chatbot is used. Since the proposed approach does not depend on labeled data, it can also be used in automated communication systems where conversations occur in near real-time.

This process resulted in a fine-tuned emotion vocabulary used by patients which was then leveraged by the NLP engine for the detection of emotion expression, taking into account negation of emotions and the quantification of emotion intensity based on intensifiers present in the text and the tense of the text.¹ For example, the NLP engine will decrease the intensity of emotions expressed about past activities.

3.1.2. Emotion state transitions using Markov chains

Research focusing on modeling emotions using Finite State Machines (FSM) possesses the potential of demonstrating emotion state changes [43–45]. In this study, based on the Markov chains, we model the emotion state changes which are represented using probabilistic automaton. In order to create the emotion transition profile, first the emotion sequence is extracted based on the chronologically ordered posts of an individual. Each post’s emotions are extracted and used to generate a sequence of emotions over time. We defined the emotion state space as; $Q = \{q_i \mid i \in [1, n]\}$ where n is the number of emotional states. The changes of the state are called *transitions* and the probabilities associated with such state changes are known as *transition probabilities*. Based on the emotion sequence of each individual, the interactions between the emotion states are modeled using a

¹ Code repository of the fundamental emotion extraction: <https://github.com/CDAC-lab/nlp-emotion-analysis-core>. Due to the patient privacy and confidentiality regulations governing this project, additional customization of the codebase, emotion ontology and clinical templates cannot be made public.

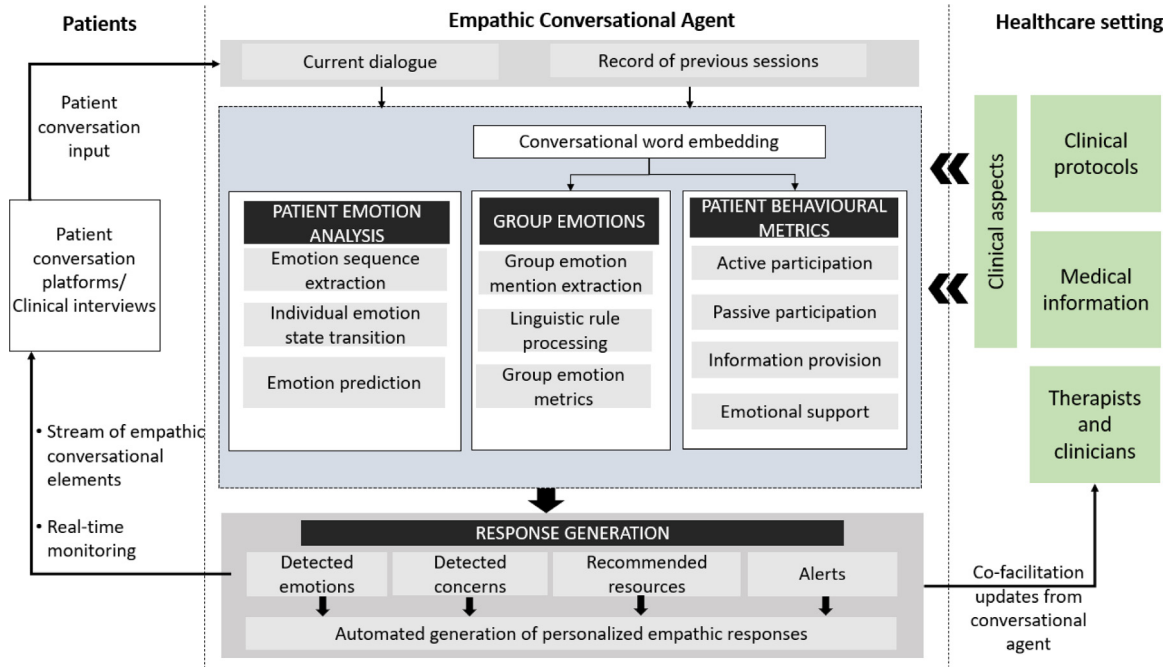


Fig. 1. Empathic conversational agent framework for real-time monitoring and co-facilitation of patient-centered care.

matrix of emotion states transferring frequencies (A_{ij}) as shown in (1).

$$A_{ij} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \quad (1)$$

where n represents the number of emotional states, a_{ij} refers to emotion transition from Q_i to Q_j where,

$$a_{ij} \in [0, 1], i, j \in [1, n] \quad (2)$$

The probabilities of outward state transitions add up to a total of 1 in each state. Emotion transitions with high propensity would represent a high transition probability indicating a higher chance for an individual to transition into the specific emotion state.

Applications of this transition model have been proposed and established in various domains [32,46]. We adopt this established approach to represent emotions modeling and prediction in a multi-user, real-time healthcare setting to automate communication with real features to inform emotion characteristics of patient behaviors. Conversation elements by each patient have been accumulated over time and their emotions have been extracted to model the emotion transition matrix which provides the probabilities of changing emotion states of each individual based on previous activity. Each emotion expression is captured, and the transition model will be automatically updated over time. This will lead to predicting the next emotion state based on previous state change probability and detecting individual emotional characteristics for each patient. These will denote individual emotion characteristics of patients over time. Patients with higher probabilities to transition into negative emotion states can be identified and therapists could provide more attention to the identified individuals.

3.1.3. Emotion prediction

Based on an individual's emotion state transitions and the time-based emotion sequence, it is possible to predict the next emotion state based on the current state. The standard Markov

property assumes that the next state is dependent on only the current state. However, by building n th order Markov models, it is possible to incorporate memory into the predictions. The assumption of the n th order Markov model is given in (3). The proposed emotion prediction model uses a second-order Markov model, where the prediction is made based on the previous two emotion states of the individual (Fig. 2).

$$P(x_i | x_{i-1}, x_{i-2}, \dots, x_1) = P(x_i | x_{i-1}, \dots, x_{i-n}) \quad (3)$$

The importance of individualized emotion models is that emotion modeling and prediction happens at an individual level, customized to each patient which in turn enhances the user experience and effectiveness of the framework. Using the prediction capabilities, it is possible to forecast the next emotion state which is highly useful for understanding patient emotion changes. In patient-centered healthcare, automated communication would have numerous benefits by adopting such a cognitive model as the smart devices will be able to make proactive decisions based on the patient's emotional state and generate personalized responses based on the emotional characteristics of each individual.

3.2. Group emotions

Group emotions and collaboration represent the extent to which a group of people progresses together through shared contextual and emotional information [47]. This behavior is apparent in online communities where people with similar interests get together to discuss their concerns and share the emotional burden. Emotions expressed within an online group environment promote good communication especially in healthcare settings [48].

The group emotion mention extraction is based on terms that are often used to express empowerment, support and collaboration such as 'give a group hug', 'we are there for you' etc. An annotated dataset of such expressions was curated the group of clinicians. They annotated 1334 posts with binary labels (YES, NO) indicating the mentions of group emotions similar to above mentioned phrases. The dataset was then split into training dataset

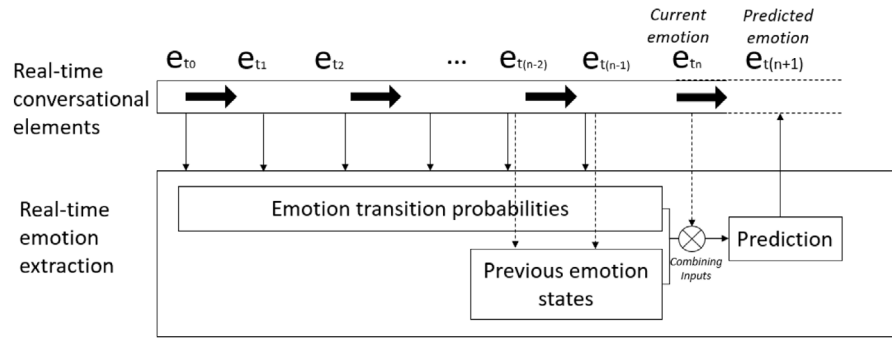


Fig. 2. Emotion prediction based on previous emotions.

(70%) and a test dataset (30%). Before training the classifier, a series of feature engineering steps were followed. Feature engineering is the process to create features by extracting information from the data. For this purpose, TF-IDF (term frequency-inverse document frequency) approach [49]. TF-IDF is a statistical measure that evaluates how relevant a word is to a document in a collection of documents. It is performed by multiplying the term frequency and inverse document frequency of the word across a set of documents. In this classification task, the TF-IDF vectorizer was used with a limit of 5000 words capturing both unigrams (single words) bi-grams (two words that occur together) as well. Next, the vectorizer was applied to the pre-processed training dataset.

Once the dataset is transformed, it was used to train multiple classifiers; Naïve Bayes, Random Forest, Support Vector Machine (SVM), Multi-Layer Perceptron (MLP) and Logistic regression models. The objective of training multiple classifiers was to increase the performance of the final classification by incorporating multiple high-performing classifiers. This technique is called 'blending', which is an ensemble machine learning technique that combines predictions from multiple models. An annotated dataset of such expressions from a clinical context was used to train a word embedding model that expands the list of expressions based on synonyms and similar contextual representations. Once this enriched vocabulary was formed, linguistic rules were applied to fine-tune the extraction. This included negation and pro-noun handling to understand the subject of the conversation, question form detection and empathetic statement detection. Empathetic statements were captured using a pre-defined template set which was formed based on the expanded list of clinical expressions. The pre-defined templates contained a selection of curated conversation points such as, 'Do you want to talk about it more?', 'You seem to be feeling down' that have been effectively used in past conversations.

Using this approach, each conversational element was classified as a binary variable indicating whether the post includes group emotion mentions or not.

The following Eq. (4) shows the score calculation which is a relative measure for each patient based on their number of conversational elements in the given time interval.

$$\beta = \frac{|C_{(t,t+T)}|}{|A_{(t,t+T)}|} \quad (4)$$

where:

β = Group emotion score

T = Specified time interval (30 min)

$A_{(t,t+T)}$ = Set of all posts occurred during the time t to t+T

$C_{(t,t+T)}$ = Set of group emotion mentioned posts occurred during the time t to t+T

3.3. Patient behavioral metrics

In patient-focused conversation groups, the amount of information shared by patients is an important measure to indicate their engagement. Especially in a healthcare chatbot application, it is pertinent to capture this information for an improved understanding of the individual. In this framework, we capture both the information and participation metrics of patients. First, in order to derive the 'emotion engagement score' of each patient, we use emotions, group emotion mentions captured by emotion extraction components. Besides, we capture specific concerns expressed by patients based on the domain and theme of discussion. As a baseline version, we used a clinical ontology of patient concerns provided by the clinicians and therapists. This included several physical symptoms and social concerns often faced by patients. The concern list was also further enriched by using the trained word2vec models which identified similar expressions used by people to express different concerns. Following these extraction steps, the relative number of high emotional posts, group emotion and concern mention posts by each patient were used to derive the 'emotion engagement score' metric.

Second, in order to capture the 'participation score' we calculated the volume of content shared by the patient using the average length of posts and the average number of posts within a conversational setting. Based on these data, we introduce a quantified measure to assess patient behaviors that encompass the information and participation score of each patient to represent the volume of content shared by the individual. The two measures were combined using the fuzzy integral. The scores for each metric were combined using a fuzzy measure to derive the final patient behavioral score.

The two metrics (participation and emotion engagement) are combined to form a score of 'patient behavioral score' using the fuzzy integral. The fuzzy integral considers the objective evidence supplied for each information source and the expected worth of each subset of information sources via a fuzzy measure. Fuzzy measures of information worth ($g(\beta)$, $g(\phi)$) are user-defined. Sugeno λ measure is calculated using Eq. (5), while the $g(\{\beta, \phi\})$ value is calculated based on the derived λ is calculated as shown in Eq. (6).

$$\lambda + 1 = \prod_{i=1}^2 (1 + \lambda_{g^i}) \quad (5)$$

$$g_{\lambda}(\{\beta, \phi\}) = g_{\lambda}(\beta) + g_{\lambda}(\phi) + \lambda [g_{\lambda}(\beta) \cdot g_{\lambda}(\phi)] \quad (6)$$

With the derived values, the overall patient behavioral metric is calculated using the following fuzzy integral as shown in Eq. (7),

$$\int_s h \cdot g = \bigvee_{i=1}^2 (h(x_i) \vee g(A_i)) \quad (7)$$

where,

$h(x)$ = value of the information source (i.e. emotion engagement and participation metrics)

A_i = ordered set of information sources

g = fuzzy measure of information worth

3.4. Response generation

Besides the emotion and behavioral metrics extraction capabilities, an architectural contribution of this framework is the intelligent message filtering approach for the propagation of empathic conversational elements for the chatbot interface and co-facilitation updates for the human therapist or care provider. The automated response generation is triggered based on multiple factors of the patient behavior such as a high probability to transition into negative emotions and prediction of imminent negative emotions. The individual's emotion transition matrix and the prediction capabilities are used to generate these notifications.

For this purpose, we have built a rule-based emotional message generation based on the 'negativity threshold' (λ) of each patient. We define the negativity threshold by calculating the average probability of transitions ending in a negative emotion state. The negative emotions are a subset of the eight emotions defined as $N = \{\text{'sadness'}, \text{'anger'}, \text{'disgust'}, \text{'fear'}\}$. The negativity threshold is calculated using the following Eq. (8).

$$\lambda = \frac{\sum_{i=1}^{|N|} \sum_{j=1}^{|Q|} P(\text{emo}_j \rightarrow \text{emo}_i)}{|N| \times |Q|} \quad (8)$$

We use λ of each patient, as an indication of the overall negativity. Based on this, we defined a rule to embed an empathetic message to the chatbot's response, if/when the negativity threshold exceeds 0.5. The empathetic response generation triggering rule (ϕ) is stated below (9).

$$\phi = \begin{cases} 1 & \lambda > 0.5 \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

Once the rule was triggered, empathetic responses are generated based on pre-defined template set (such as 'Do you want to talk about it more?', 'You seem to be feeling down') that acknowledge the patient's current emotion state. This will personalize the communication compared to generic question-answer type conversation flow. A completely automated empathetic response generation is outlined as future work of the study. In addition to the customized response, alerts will be generated to the therapists regarding the negativity threshold, so human intervention is made to support the patients in distress.

3.4.1. Resource recommendation

In order to customize the patient's experience when using this framework, we have introduced the resource recommendation capability. This functionality suggests potential resources for further reading based on the concerns expressed by the patient. The concerns are extracted using the approach mentioned in 3.3 using a defined set of medical and social concerns with the use of the trained word embedding models. Once a patient expresses concern, the information is recorded, and the framework will use a set of clinical resources related to the concern and embed that information in the response message to the patient. This automated step personalizes the communication between the patient and the PCC setting as well as provides individualized care and attention without therapist intervention.

The functionality of the proposed framework in a PCC setting can be summarized as follows. Initially, patient emotions, intensities, and emotion state changes which indicate the overall progression of the emotion behaviors are detected and aggregated. Individuals identified as demonstrating a higher propensity to

transit to negative emotion states based on the transition matrix and predictions of imminent negative emotion are notified to the healthcare practitioners as well as used for automated response generation. Next, the conversations are analyzed to extract group emotion mentions and concerns in order to better understand patient behaviors and based on this understanding recommend relevant resources. These insights are then combined to derive patient behavioral score to develop a further understanding of the overall engagement and behavioral changes of all patients. Furthermore, the visualizations generated via the emotion transition behaviors enable the visual construction of free-flowing patient conversations depicting the emotional changes. These technical capabilities and methodological innovation embedded in a chatbot framework enable real-time monitoring of patient conversations, automated response generation and co-facilitation in a PCC setting. The benefits to a healthcare system include individualized emotional support to patients, improved mental health and wellbeing, and reduced workloads for healthcare practitioners through the automation of first-level communicative labor.

4. Evaluation and validation of the proposed framework

The evaluation of the framework was conducted using two separate datasets. First, the capabilities of the framework to detect emotions, group emotions, resource recommendation and the ability to visualize the transitions were investigated using a clinical protocol designed for its application in an online support group setting for cancer patients and caregivers across six provinces in Canada [15,43]. For this purpose, patient conversations of real-time support group were used. Next, a conversation dataset from Kaggle [50] was used to evaluate the emotion transition prediction capabilities that serves as a sanity check. Table 1 presents the overview of the two datasets used in this study.

4.1. Evaluation of the framework using online cancer support group dataset – Cancer Chat Canada

Cancer Chat Canada (CCC) [51] offers professionally led text-based online support groups (OSG) to cancer patients and caregivers across Canada, with transcripts available to therapists and participants for post-session review [15,43]. The OSG in CCC is manual-based, with session-specific themes, readings and interactive activities. Therapists lead group discussions based on the manual, address new topics brought up by participants during the session, attend to an individual's emotional concerns, and facilitate mutual support.

For the experiments, we used over 120,000 conversations by 320 patients which were accumulated over multiple sessions in the 2016–2017 time period. These 120,000 conversations were used to train the two separate word embedding models using single tokens and bigrams. This enabled to capture the contextual information related to cancer support groups in Canada and identify specific expressions of emotions and concerns by patients. The series of experiments demonstrated below express the potentiality of embedding these functionalities into chatbot systems as well as aids the visualization of insights generated using unstructured clinical conversations. The validation of the outcomes was conducted by a group of clinicians and therapists.

4.1.1. Emotion detection and transitions of patients

The emotion detection using the trained word embedding models, enabled classifying each post with a specific emotion category. Each post was labeled with one or many emotion categories based on the emotions expressed in the post. This enabled calculating the emotion intensities for each patient as well as for

Table 1
The datasets used in the experiments.

Dataset	Volume
1. Patient conversations from Cancer Chat Canada platform (not publicly available due to confidentiality and privacy)	120,000 conversations by 320 patients
2. User dialog dataset from Kaggle (public)	3,000,000 messages by 702,669 (average 5 messages per user)

Table 2
Performance of classifiers used to detect group emotion statements.

Classifier	Precision	Recall	F1 score
SVM	0.52	0.8	0.63
Naïve Bayes	0.79	0.79	0.79
MLP	0.78	0.77	0.77
Random forest	0.74	0.77	0.72
Logistic regression	0.92	0.74	0.82

the group within a support group session. The classified emotions were evaluated by a group of clinicians and therapists for validation. They manually investigated the correctly classified and incorrectly classified posts with emotions, that yielded an average F1 score of 0.7 for emotion extraction. The incorrectly classified posts often contained ambiguous emotional expressions related to idioms, indirect expressions or reference to other people's experiences. Once the F1 score of 0.7 was achieved, emotion transition profiles were created for each patient based on their conversations.

This created an emotion print of each patient resembling their individual emotional characteristics and behaviors. Fig. 3 represents such emotion transition matrices of three individuals with the probabilities of their emotion transitions who participated in the same chat session. The difference in emotion transitions infers different moods and emotion states of individuals that will be essential for emotional behavior monitoring using automated systems.

In addition, another experiment was carried out to capture temporal transitions over different time intervals during a single session. The conversations were split into 30-minute intervals and the emotion extraction and state transition models were created separately for the time intervals. This resulted in investigating the emotion transition and propagation of an individual within a session at a granular level (Fig. 4). This further facilitates the visualization of clinical conversations over time which can be applied to any other format of direct engagement such as interviews, conversations and therapy sessions.

4.1.2. Group emotions within a session

In order to evaluate the group emotions, several classifiers were trained to classify each post indicating the presence of group emotions. Table 2 shows the performance of each classifier when evaluated on the test dataset.

Based on this, the top three classifiers were selected; Naïve Bayes, MLP and Logistic Regression. The outcomes of all three classifiers were used to make the final prediction. If a post is classified into the same label by 2/3 classifiers, then the output label was used as the final outcome. A confidence value was generated for each classification based on the weighted f1 scores of each classifier. The average F1 score using three classifiers was 0.8.

This method was used to investigate the group emotions within a session, in which we calculated the group emotion score at 10-minute intervals. This was an aggregated score to measure the collaboration within a session. The implementation of this functionality enables the chatbot to understand the group

Table 3
Snapshot of concerns vocabulary created using the trained word embedding models.

Concern identified by the therapist	Enriched terms using the word embedding models
Grief and Loss	{death, losses, loss, passed away, trauma}
Fatigue	{stress, discomfort, dizziness, tiredness}
Anxiety	{panic attacks, breathing, fear, panic}
Caregiver support	{supportive, caregivers, caregiving}
Finance	{income, cost, insurance}
Distress	{stress, helplessness, moody, hopeless}

atmosphere, the collaboration and the support provided by patients with themselves. Identification of low group collaboration and engagement by the chatbot will enhance the co-facilitation capabilities as well as make the chatbot more intelligent and receptive to the changes occurring in the virtual environment.

Fig. 5 demonstrates the group emotion scores overtime in two separate sessions. It can be observed that session 1 is comparatively low when compared to session 2. Such differentiation could be automatically identified by setting a threshold value to generate notifications so that whenever the group collaboration and emotions fall below the threshold, an automated response can be generated through the chatbot interface.

4.1.3. Patient behavioral metrics of participants

The patient behavioral score denotes the overall engagement of patients within a PCC setting. It is expected that an effective PCC setting facilitates information sharing, experience sharing and patient engagement in the healthcare process. To this end, it is important to detect individuals with low engagement which calls for therapist intervention. Fig. 6 demonstrates the patient behavioral scores captured by the co-facilitator chatbot framework. The patient behavioral score is captured at 10-minute intervals enabling a time-based analysis of participant's engagement. It can be seen that participant 1 has a low engagement in the first half of the session, but in the latter part, more content has been shared. In contrast, participant 4 showcases an increased engagement in the first half of the session but gradually declines over time.

4.1.4. Resource recommendation

The resource recommendation functionality was implemented using the outcomes of emotion extraction (Section 3.1) and concern extraction (Section 3.3). Using the word embedding models trained on the CCC conversation data, the concerns were augmented to enrich the extraction. Table 3 shows the capability of this enrichment.

In this functionality, each conversation was analyzed to extract mentions of the afore-mentioned concerns. If there are similar expressions mentioned, then it was recorded on the patient's profile. Once such concern was identified the relevant resources will be suggested to the patient. This process was conducted on the conversation dataset and Table 4 demonstrates the results of the resource recommender functionality. The relevant resources were fed into the system by domain experts and therapists which can be updated frequently. The chatbot framework is able to detect

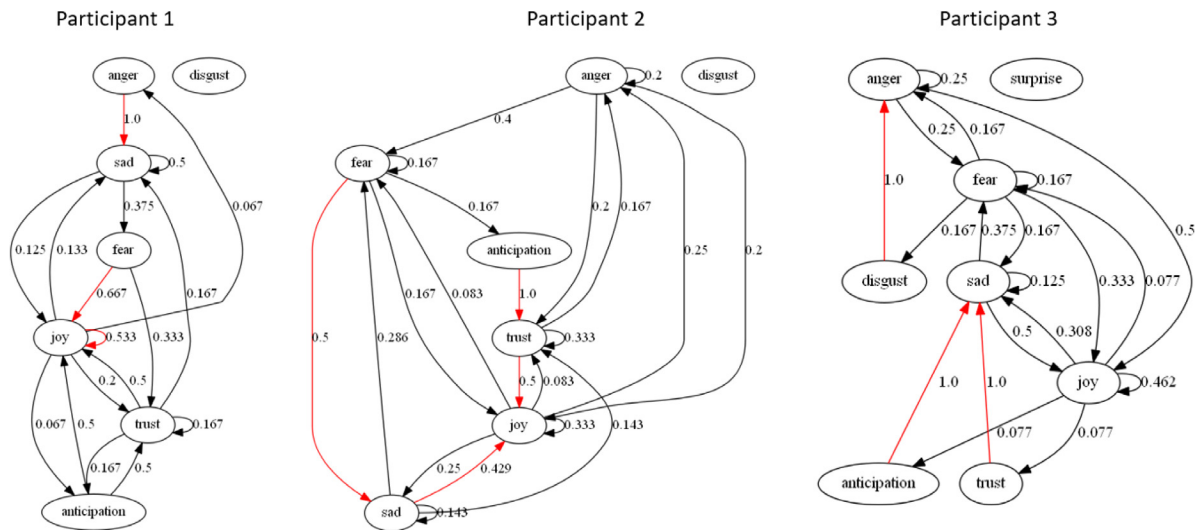


Fig. 3. Emotion transitions of three individuals during a single session.

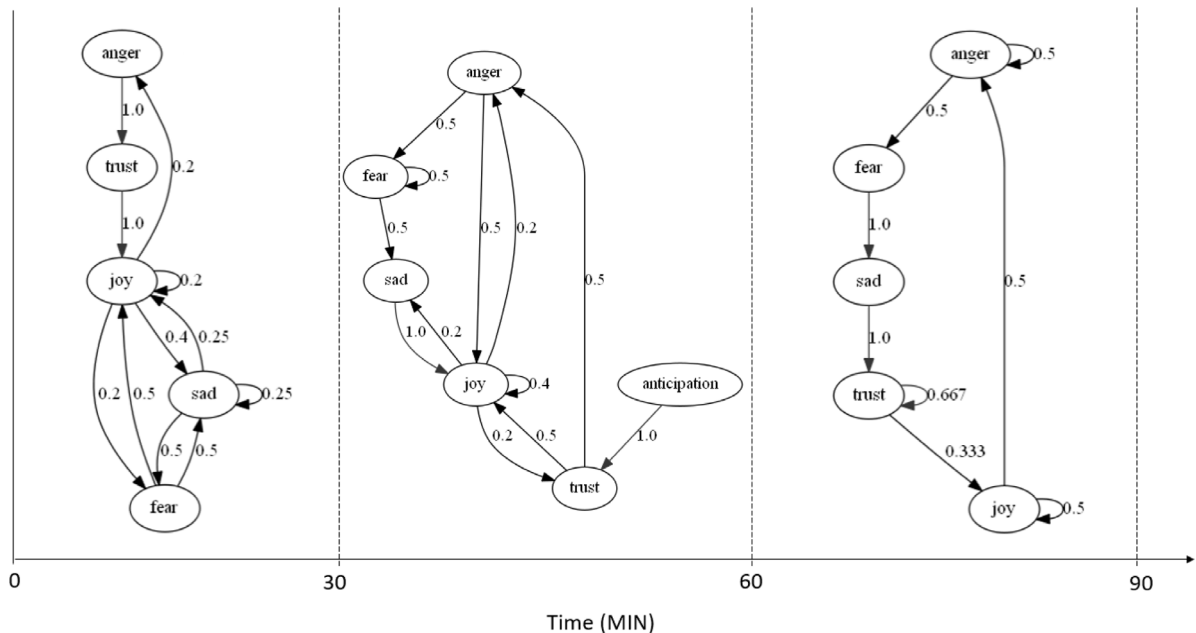


Fig. 4. Temporal emotion state changes of an individual within a session.

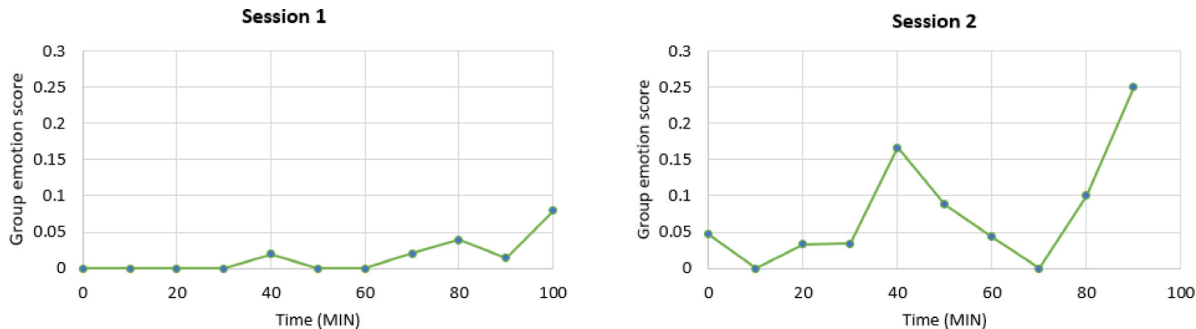


Fig. 5. Group emotion changes over time within a session.

the concern, relevant posts mentioning the concern and suggest relevant resources for each participant in an automated process. This ability enhances the context-awareness of the chatbot as well as promotes personalized care for patients.

Once this feature was implemented, a domain evaluation was conducted based on expertise knowledge from the group of clinicians and therapists. In this process, for each post in the validation dataset, the automatically generated resource was checked for

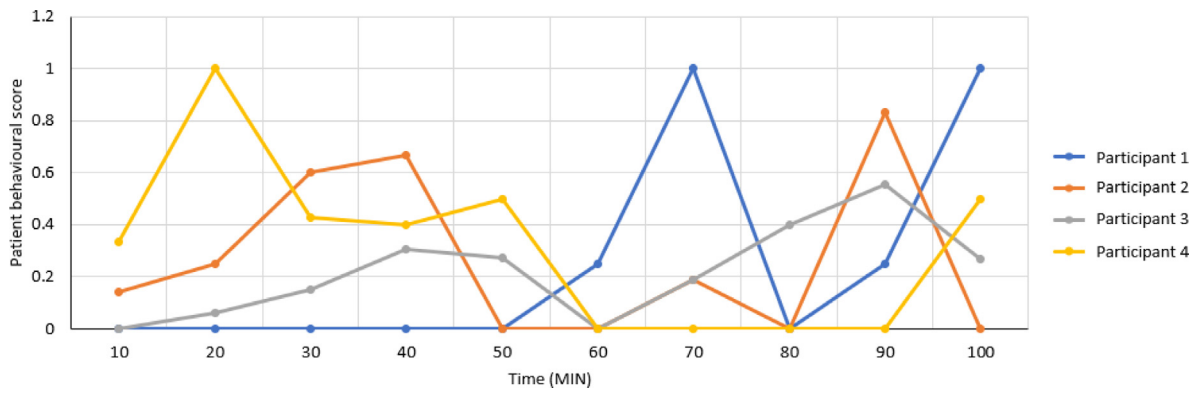


Fig. 6. Patient behavioural metrics during a session.

Table 4

Sample outcomes from the resource recommenders.

Participant ID	Detected concern	Related posts	Recommended resources
Participant 1	Grief and loss	“... my brother in law died from cancer 3 weeks ago and I'm still trying to get over it”, “grief doesn't go away easily”	“https://livingmyculture.ca/topic/after-death-and-ceremonies/”, “https://kidsgrief.ca/”, “https://support.desouzainstitute.com/kb/article/54-nucare-manual”
Participant 2	Depression	“true so much waiting is depressing and not easy”	“http://www.bccancer.bc.ca/health-info/coping-with-cancer/emotional-support/sadness-and-depression”, “https://support.desouzainstitute.com/kb/article/54-nucare-manual”, “https://www.cancercareontario.ca/en/symptom-management/3986t”, “https://www.cancersupportcommunity.org/cancer-support-helpline”
Participant 3	Caregiver Support	“I was a caregiver to my older brother for two years and still struggling ”, “I think being a caregiver is harder than being the patient”	“https://ontariocaregiver.ca/find-support/helpline/”, “https://www.cancersupportcommunity.org/”, “https://www.cancersupportcommunity.org/”

accuracy. A correct classification was where the recommended resources by the system matched the concern expressed in the post. The accuracy was determined by the team of therapists. A confusion matrix was generated based on the correct and incorrect classifications of the resources. This yielded a F1 score of 0.87.

4.1.5. Customized response generation and co-facilitator updates

Once the conversations are investigated for emotional behaviors, group emotions and patient behavioral metrics, these outcomes are used to generate the response and real-time alerts and warnings. Identification of triggered events is used to customize the chatbot's response to patients. In the proposed framework, we have integrated a rule-based empathetic response generation based on a pre-defined collection of empathetic messages. Based on this information the chatbot provides a customized response embedded with relevant resources and an empathetic statement to the participant. In addition to the customized response, the chatbot provides updates to the therapist establishing the roles as a co-facilitator. This enables the integration of such a chatbot framework into existing patient-centered healthcare settings where automated communication is required. In addition, these capabilities can be embedded in to existing chatbot applications to support emotion awareness and patient behavior analysis in a real-time manner, as an adjunct to human involvement.

4.2. Evaluation of emotion prediction

A further evaluation was conducted as a sanity check to evaluate the performance of the emotion prediction using the emotion

transition models. For this purpose, we used a Kaggle dataset that contains approximately three million dialogs from 702,669 participants due to the high volume of data that was not available in the previous cancer chat dataset. Each participant's conversational elements were accumulated and sorted according to the time of occurrence in order to generate a series of input by each participant over time which enabled analyzing long-term emotion transition and prediction. In addition to the basic emotions, we have also added 'stress' and 'frustration' to the emotion extraction process thereby deriving a sequence of emotions for each participant's conversation thread. Using this information an experiment was conducted to evaluate the feasibility of using emotion state transition and future emotion state prediction capabilities on real-world dialogue datasets.

Each participant's conversational elements were ordered chronologically and split into 70:30 partitions. The first 70% of conversational elements were taken to generate the emotion state transition matrix (training), while the remaining 30% of conversational elements were taken for validating the prediction results. The prediction was made based on the 2nd order Markov model, where the previous two emotion states were taken to make the prediction about the next emotion state. This process was carried out for all the participants and an average measure was taken to calculate the accuracy of the prediction model, providing a more generalized and valid measure of model performance.

It was observed that in 79% instances, the model was able to correctly predict the next emotion as positive or negative, and in 63% instances, the correct emotion out of all emotions was predicted. In order to assess the prediction capabilities with more

input data, separate prediction experiments were carried out with participants with more than 20 and 30 posts. This resulted in accuracies of 67.21% and 72.83% for the 20 input post group and 30 input post group respectively. This demonstrated that the prediction accuracy increases with the availability of more historic data. This empirical evaluation of the emotion behavior capturing using a benchmark dataset confirms its alignment with the state-of-the-art in individualized conversational NLP research and its capacity to handle large-scale unstructured data, which is commonplace in a healthcare setting.

5. Conclusion

In this paper, we have proposed an empathic conversational agent framework based on an ensemble of natural language processing techniques and artificial intelligence algorithms for real-time monitoring and co-facilitation of patient-centered healthcare for improved mental health and wellbeing outcomes. The ensemble of AI algorithms and NLP techniques detect and analyze patient emotions, emotion transitions, group emotions, and patient behavioral metrics. This ensemble was adapted for multi-user settings from the emotion transition models used to represent an individual's emotional status [32,46]. We have demonstrated the effectiveness and practical value of the framework and its core capabilities by empirically evaluating on a benchmark dataset and further validation based on a clinical protocol designed for its real-world application on an online support group for cancer patients. The outcomes of these experiments confirm the suitability of the proposed framework as an adjunct to healthcare practitioner involvement in a patient-centered healthcare setting, leading to improved mental health and wellbeing outcomes.

We acknowledge that emotion extraction can be improved by addressing the current limitations related to detecting ambiguous expressions, idioms and indirect emotion statements [52,53]. Furthermore, advanced NLP such as transformer-based language models, will be used to improve the contextual understanding of the patient conversations. This will potentially enhance the automated patient communication and further minimize the intervention by therapists and clinicians.

CRediT authorship contribution statement

Achini Adikari: Conceptualization, Formal analysis, Methodology, Writing – original draft. **Daswin de Silva:** Conceptualization, Formal analysis, Methodology, Writing – original draft. **Harsha Moraliyage:** Formal analysis, Methodology, Writing – review & editing. **Damminda Alahakoon:** Conceptualization, Formal analysis, Methodology, Writing – review & editing. **Jiahui Wong:** Conceptualization, Methodology, Investigation, Project administration. **Mathew Gancarz:** Conceptualization, Data curation, Methodology, Project administration. **Suja Chackochan:** Data curation, Software, Visualization. **Bomi Park:** Data curation, Investigation, Validation. **Rachel Heo:** Data curation, Investigation, Validation. **Yvonne Leung:** Conceptualization, Methodology, Investigation, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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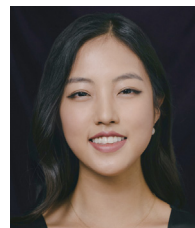
utilized by UHN Researchers. He holds a Masters of Management Science (MMSc) from the University of Waterloo and has completed a certificate in the Management of Enterprise Data Analytics from the University of Toronto as well as Microsoft, Apple and Cisco certifications.



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Yvonne Leung, Scientist at de Souza Institute, Clinical Investigator at the Princess Margaret Cancer Centre, University Health Network, and Assistant Professor in Psychiatry, University of Toronto. Dr. Leung explores the application of deeplearning based natural language processing algorithms to provide e-monitoring for therapist-led online support group participants' functional and emotional outcomes in real-time, while accessing group coherence, group climate, and detecting issues (e.g. mental health deterioration) that are amendable to interventions.